

INTERNSHIP OUTLINE FOR BACHELOR/ MASTER STUDENT

Department: Department of Information and Communication Technology

Topic: A Hybrid Framework for 3D Edge Detection Combining Geometric Features and Deep Learning

Description:

-This project addresses the challenge of identifying structural boundaries and geometric discontinuities within unorganized 3D point clouds. Traditional edge detection often fails in noisy or low-density areas, while pure deep learning models can be computationally expensive and lack geometric precision.

-This research proposes a hybrid framework:

- + Geometric Stream: Uses mathematical descriptors (e.g., PCA-based surface variation or Angular Gaps) to find "candidate" edge points.
- + Deep Learning Stream: Uses a lightweight Point-based Neural Network (like PointNet++ or a simplified DG CNN) to validate these candidates and filter out sensor noise.

The project will use the House3D dataset to train and validate the model in a controlled, indoor environment.

This internship requires the following skills:

- 3D Computer Vision
- Programming & Frameworks
- Mathematical Foundations
- Data Management

Objectives:

- To implement a geometric edge detection algorithm
- To develop a hybrid framework that improves detection accuracy by combining geometric and deep learning features.

The tasks include:

- Environment Configuration: Setting up a workspace with Python, Open3D, and CUDA support.
- Data Preparation: Processing the Microsoft 7-Scenes and House3D datasets for testing.
- Model Integration and Evaluation

Place, Date.....

Student

(Full-name and signature)

Khanh

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